

The Four Pillars of Computational Thinking

The thinking you do before you write a single line of code.

1. DECOMPOSITION

Break a problem too big to solve into pieces small enough to solve.

CivicTrack: "handle a service request" becomes take it in, route it, track it, close it.

2. PATTERN RECOGNITION

Notice what repeats, so you only have to solve it once.

CivicTrack: a pothole, a broken light and a water leak are all just "a report with a location".

3. ABSTRACTION

Ignore the details that do not matter for the problem in front of you.

CivicTrack: to route a request you need its category. You do not need the citizen's life story.

4. ALGORITHM DESIGN

Write the steps in an order that works every time, not just this time.

CivicTrack: the exact rule that decides which department a new request is sent to.

None of this is typing. This is the part of programming you can practise on paper, in the shower, or in a meeting.